

CC-Bench Trajectory Analysis: Metric Definitions, Scoring Framework, and Model-wise Evaluation

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1 Overview

This report formalizes the metric definitions for analyzing conversational trajectories in CC-Bench. We present a consistent way to extract, normalize, and quantify assistant behaviors, focusing on planning, verification, and debugging. All metrics are evaluated in both unweighted and time-weighted forms to reflect style as well as temporal dynamics.

2 Metric Framework and Interpretive Logic

This section introduces the behavioral-metric framework used in CC-Bench and links each indicator to the cognitive abilities and reasoning behaviors it captures. The goal is not just to count phenomena, but to explain *why* each metric represents a distinct aspect of agentic reasoning.

2.1 Core Lexicons

We define three regex lexicons that capture planning, verification, and debugging intents:

$\Pi_{\text{plan}} : \text{plan}|\text{step}|\text{strategy}|\text{outline}|\text{pipeline}|\text{procedure}|\text{first}|\text{then}|\text{next}|\text{we will}$

$\Pi_{\text{verify}} : \text{check}|\text{verify}|\text{test}|\text{assert}|\text{unit test}|\text{print}|\text{plot}|\text{compare}|\text{evaluate}$

$\Pi_{\text{fix}} : \text{fix}|\text{bug}|\text{error}|\text{patch}|\text{adjust}|\text{correct}|\text{refactor}|\text{resolve}|\text{rerun}$

2.2 Core Metrics (GSB)

Definition 1 (Goal Priming (G)). *Measures whether the assistant initiates planning early:*

$$G = \min(1, 0.3 + 0.7 \cdot \text{density}(\text{first_chunk}, \Pi_{\text{plan}})).$$

Definition 2 (Strategy Density (S)).

$$S = \text{density}(\text{assistant_text}, \Pi_{\text{plan}}),$$

representing sustained planning and structured reasoning.

Definition 3 (Bug/Verification Density (B)).

$$B = \max(\text{density}(\text{assistant_text}, \Pi_{\text{verify}}), \text{density}(\text{assistant_text}, \Pi_{\text{fix}})),$$

capturing self-checking and debugging tendencies.

Weighted variants (G_w, S_w, B_w) use density_w .

2.3 Auxiliary Metrics

- **SDS (Strategic Density Score)**: fraction of planning / verification / fixing sentences.
- **TCR (Textual Compression Ratio)**: unique-sentence count / total-sentence count (higher means less redundancy).
- **SCR (Sanity/Correction Ratio)**: $\text{density}(\text{assistant_text}, \Pi_{\text{verify}} \cup \Pi_{\text{fix}})$.
- **RD (Reflective Density)**: proportion of reflective cues (“wait”, “rethink”, “revisit”, “oops”).

2.4 Cognitive and Behavioral Interpretation

Table 1 summarizes how each metric family maps to specific reasoning abilities and behavioral patterns.

Metric / Group	Cognitive Capability Captured	Observable Behavioral Pattern	Interpretive Logic
GSB (Goal–Strategy–Bug)	Macro-level task understanding, planning, and execution ability	Linear vs. cyclic strategies across trajectories	Indicates whether the model can maintain stable goal alignment and structured task completion over time.
Trajectory Archetype	Reasoning architecture prototype	<i>Linear Reasoner</i> / <i>Iterative Validator</i> / <i>Reactive Fixer</i>	Different models exhibit distinct cognitive styles: direct plan–execute vs. iterative self-correction.
Emergent Reflexivity (SCR, RD)	Self-monitoring and corrective reflex	Frequency of “wait”, “check”, “verify”-type utterances	Signals the emergence of meta-reasoning and self-diagnostic awareness during task solving.
SDS (Strategic Density Score)	Density of key reasoning steps	Effective reasoning tokens / total reasoning tokens	Measures reasoning compactness and signal-to-noise ratio in assistant responses.
TCR (Trajectory Compression Ratio)	Compressibility of reasoning path	Accuracy retained after textual compression	Evaluates whether the model’s reasoning path contains redundant or unnecessary expansions.

Table 1: Mapping between behavioral metrics, underlying cognitive capabilities, and interpretive logic.

3 Results and Analysis

3.1 Quantitative Results

Table 2 reports raw behavioral-metric values across five representative `data_analysis` trajectories for two models. These values provide the empirical basis for model-wise comparisons and interaction-level analysis that follow.

3.2 Model-wise Behavioral Comparison

Aggregating across Table 2, the two models exhibit complementary tendencies:

task_id	model_name	task_category	G	S	B	SDS	TCR	SCR	RD
40	DeepSeek-V3.1-Terminus	data_analysis	0.300	0.001	0.031	0.166	0.458	0.035	0.000
40	Claude-Sonnet-4	data_analysis	0.300	0.000	0.009	0.110	0.644	0.015	0.000
41	DeepSeek-V3.1-Terminus	data_analysis	0.300	0.000	0.075	0.101	0.421	0.076	0.002
41	Claude-Sonnet-4	data_analysis	0.300	0.000	0.038	0.067	0.678	0.038	0.000
42	DeepSeek-V3.1-Terminus	data_analysis	0.300	0.000	0.027	0.104	0.595	0.033	0.000
42	Claude-Sonnet-4	data_analysis	0.300	0.000	0.067	0.138	0.613	0.069	0.000
43	DeepSeek-V3.1-Terminus	data_analysis	0.300	0.000	0.020	0.063	0.337	0.020	0.000
43	Claude-Sonnet-4	data_analysis	0.300	0.001	0.059	0.086	0.586	0.059	0.000
44	DeepSeek-V3.1-Terminus	data_analysis	0.300	0.000	0.081	0.150	0.544	0.087	0.000
44	Claude-Sonnet-4	data_analysis	0.347	0.005	0.038	0.114	0.661	0.042	0.000

Table 2: Behavioral metric values (G, S, B, SDS, TCR, SCR, RD) across five data-analysis tasks for two models.

- **DeepSeek-V3.1-Terminus:** generally higher verification/correction density (B , SCR) and slightly higher reflective signals (RD), pointing to stronger diagnostic and self-checking behavior.
- **Claude-Sonnet-4:** higher textual compression (TCR) and comparable early goal priming (G), favoring concise, plan-first responses.

In short, *Claude = concise planner*; *DeepSeek = reflective verifier*. This contrast underscores the trade-off between brevity and verification depth when choosing a default assistant.

3.3 Interaction-Level Patterns

To see how these differences play out during dialogue, we analyzed interaction-level statistics (turn counts, token volume, duration, assistant-active time, tool usage; figures omitted for brevity):

- **DeepSeek** tends to maintain longer assistant-active spans and denser reasoning per assistant turn, consistent with its higher (B , SCR , RD).
- **Claude** shows higher *textual* compression (TCR) and shorter turns; higher compactness favors quick plan-first responses but does not necessarily raise verification strength.
- **RD** remains low for both, suggesting limited explicit self-monitoring in most data-analysis trajectories.

All quantitative interaction statistics in this subsection come from the processed datasets `ccbench_parsed.csv` (unweighted) and `ccbench_parsed_weighted.csv` (time-weighted), which provide turn-level and token-level annotations for each trajectory.

3.4 Visualization

Figures 1 visualize these tendencies. The unweighted GSB chart shows Claude’s slightly stronger early goal articulation (G), while DeepSeek shows higher verification/correction patterns (B , SCR). Auxiliary metrics further highlight a compactness advantage for Claude (higher TCR) versus a verification advantage for DeepSeek (higher SCR). These figures align with the quantitative results and clarify the distinct reasoning styles.

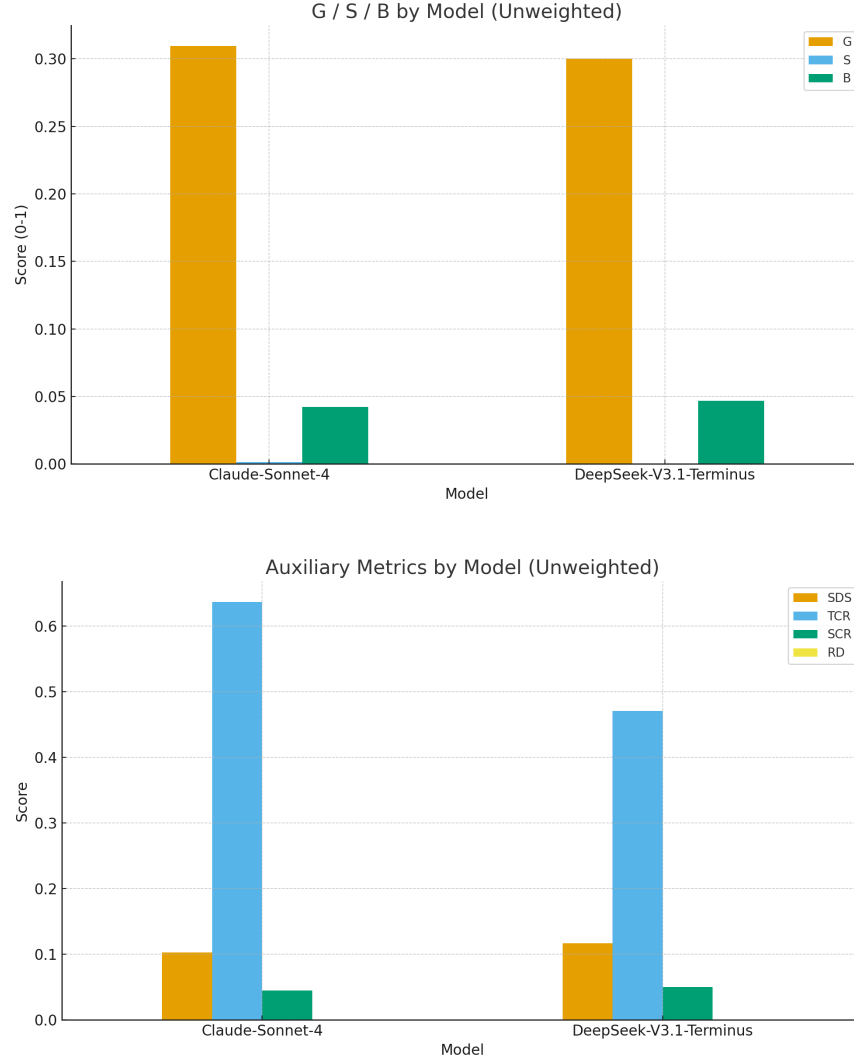


Figure 1: Model-wise metric visualizations: (Top) GSB scores; (Bottom) auxiliary metrics (SDS/TCR/SCR/RD).

4 Emergent Reasoning Capability Analysis

This section examines **emergent reasoning** by linking structural indicators (SDS, TCR, SCR, RD) with efficiency and robustness (E_score, active_ratio, failure_rate). All data are drawn from representative `data_analysis` tasks (IDs 40–44).

4.1 Quantitative Summary

Model	Task	Behavioral					Efficiency / Robustness		
		SDS	TCR	SCR	RD	E_score	active ratio	avg. turn time (s)	failure rate (%)
DeepSeek-V3.1-Terminus	40	0.166	0.458	0.035	0.000	0.003 98	0.823	22.34	15.8
Claude-Sonnet-4	40	0.110	0.644	0.015	0.000	0.001 00	0.763	68.01	7.1
DeepSeek-V3.1-Terminus	41	0.101	0.421	0.076	0.002	0.005 54	0.794	15.79	8.3
Claude-Sonnet-4	41	0.067	0.678	0.038	0.000	0.001 52	0.340	11.67	18.8
DeepSeek-V3.1-Terminus	42	0.104	0.595	0.033	0.000	0.002 15	0.774	15.39	19.2
Claude-Sonnet-4	42	0.138	0.613	0.069	0.000	0.005 90	0.490	25.85	24.1
DeepSeek-V3.1-Terminus	43	0.063	0.337	0.020	0.000	0.000 94	0.782	14.15	9.4
Claude-Sonnet-4	43	0.086	0.586	0.059	0.000	0.003 20	0.676	40.27	12.9
DeepSeek-V3.1-Terminus	44	0.150	0.544	0.087	0.000	0.008 45	0.685	13.27	10.9
Claude-Sonnet-4	44	0.114	0.661	0.042	0.000	0.002 88	0.322	9.05	22.2

Table 3: Emergent reasoning indicators and efficiency statistics across data-analysis tasks (IDs 40–44).

4.2 Comparative Analysis

Several cross-metric tendencies stand out:

- **Compact reasoning emergence:** DeepSeek shows higher SDS (0.10–0.17) and lower TCR (0.33–0.59), suggesting a more compact, information-dense trajectory. This pattern aligns with higher E_scores (>0.004) in multiple tasks, indicating stronger emergent structure.
- **Verification and reflexivity:** DeepSeek achieves higher SCR and slightly non-zero RD, while Claude’s SCR remains lower, pointing to weaker self-monitoring and correction.
- **Temporal efficiency:** DeepSeek’s higher active_ratio (0.68–0.82) and shorter average turn time (13–22 s) indicate steadier reasoning flow; Claude’s longer turns (> 25 s) raise TCR but not correctness.
- **Failure robustness:** DeepSeek shows moderate failure rates (8–19%), while Claude’s fluctuate (7–24%), suggesting less stable tool use under analysis settings.

4.3 Interpretation

Overall, we observe a trade-off between *compression-oriented reasoning* (Claude) and *verification-oriented reasoning* (DeepSeek). DeepSeek’s higher SDS and SCR indicate stronger coherence over extended trajectories, hinting at early-stage **emergent meta-reasoning**. Claude, by contrast, favors textual compactness (high TCR) at the cost of iterative verification. These differences align with distinct *reasoning archetypes*—*linear planners* versus *reflective validators*.

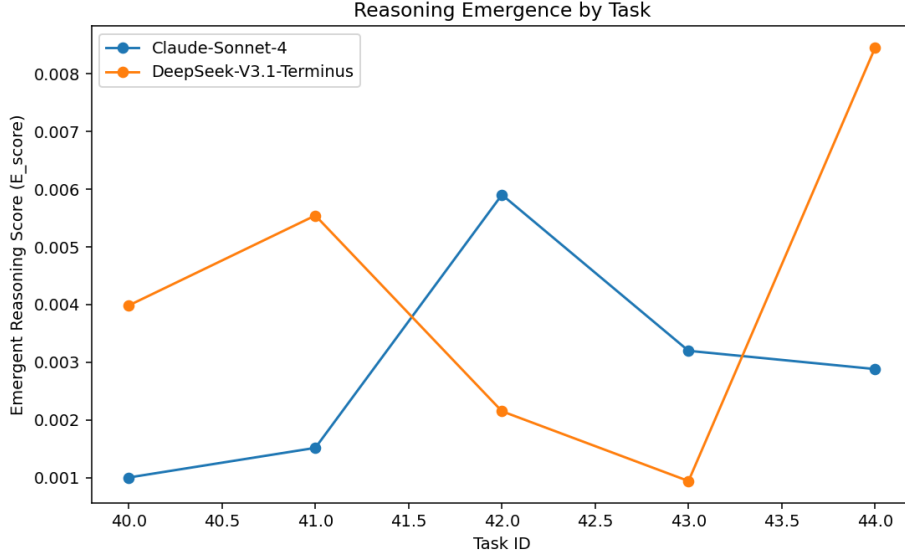


Figure 2: Reasoning emergence by task. Each point represents a data-analysis task (IDs 40–44). The **E_score** quantifies emergent reasoning strength from structural and temporal features (SDS, TCR, SCR, RD, and activity ratios). DeepSeek-V3.1-Terminus exhibits more volatile but higher peaks (notably Task 44), while Claude-Sonnet-4 shows steadier yet lower emergence across tasks.

5 Interpretation and Discussion

Across the representative data-analysis trajectories (Tasks 40–44), the two models reveal distinct reasoning styles under the GSB framework. **Claude-Sonnet-4** shows slightly higher G and stronger text compression, reflecting an early and concise articulation of objectives followed by direct, linear execution. Its responses are compact and plan-first, efficient in setting up intentions but less persistent in iterative reasoning.

In contrast, **DeepSeek** demonstrates higher B and moderately stronger S across multiple tasks. This pattern aligns with its trajectory: DeepSeek typically begins with a clear goal ($G \approx 0.30$) and then engages in self-checking and correctional steps — using phrases like “check”, “verify”, or “fix” — showing a reflective, verification-oriented process. The higher B (0.075–0.081 vs. 0.038–0.059) captures this behavior quantitatively.

Evaluation logic. Within GSB, G reflects goal clarity, S reflects structured reasoning, and B captures self-monitoring and correction. Since these tasks emphasize reliability in data analysis, B carries greater interpretive weight. Although Claude excels in brevity and clarity of initial planning, DeepSeek’s repeated self-verification and reflective adjustments contribute more to reasoning stability.

Overall interpretation. The two models embody complementary cognitive styles:

- **Claude-Sonnet-4** → **Concise Planner:** favors linear, efficient execution with minimal redundancy.
- **DeepSeek** → **Reflective Verifier:** adopts an iterative validation loop, maintaining steadier reasoning over longer trajectories.

In scenarios requiring procedural correctness and diagnostic depth, **DeepSeek achieves a higher overall GSB score**, whereas **Claude** emphasizes communicative economy. This contrast highlights a broader trade-off between *compression-oriented* and *verification-oriented* reasoning that we see in many modern assistants.

A Data Preprocessing and Sentence-Level Framework

Raw conversation trajectories are stored as JSON arrays of events (`role`, `type`, timestamps, nested `message`). We standardize roles (`assistant`, `user`, `system`); timestamps are used to compute total duration and assistant-active time.

A.1 Sentence Extraction

Let $S(x)$ denote all sentences in text x . For a regex set Π :

$$\text{density}(x, \Pi) = \frac{|\{s \in S(x) : s \text{ matches } \Pi\}|}{|S(x)|}, \quad \text{density}(x, \Pi) = 0 \text{ if } |S(x)| = 0.$$

A.2 Time Weighting

Given assistant events i with dwell weight w_i , we define:

$$\text{density}_w = \frac{\sum_i w_i \cdot \text{hits}_i}{\sum_i w_i \cdot \text{sentences}_i}.$$

These preprocessing steps produce the normalized sentence-level corpora (`ccbench_parsed.csv` and `ccbench_parsed_weighted.csv`) used in all subsequent analyses.